A Cholesky Decomposition based Massive MIMO Uplink Detector with Adaptive Interpolation

Rakesh Gangarajaiah, Hemanth Prabhu, Ove Edfors, and Liang Liu
Department of Electrical and Information Technology, Lund University, Sweden
Email: {rakesh.gangarajaiah, hemanth.prabhu, ove.edfors, liang.liu}@eit.lth.se

***Abstract*—An adaptive uplink detection scheme for a Massive MIMO (MaMi) base station serving up to 16 users is presented. Considering user distribution in a cell, selective matched filtering (MF) is proposed for non-interference limited users and a Cholesky decomposition (CD) based zero-forcing (ZF) detector is implemented for the remaining users. Channel conditions such as coherence bandwidth are exploited to lower computational complexity by interpolating CD outputs. Performance evaluations on measured MaMi channels indicate a reduction in computation count by 60 times with a less than 1 dB loss at an uncoded bit error rate of 10^{-3}. For the CD, a reconfigurable processor optimized for 8×8 matrices with block decomposition extension to support up to 16×16 matrices is presented. Circuit level optimizations in 28 nm FD-SOI resulted in an energy of 1.4 nJ/CD at 400 MHz, and post-layout simulations indicate a 50% reduction in power dissipation when operating with the proposed interpolation based detection scheme compared to traditional ZF detection.**

I. Introduction

Massive MIMO (MaMi) is a promising candidate for the next generation wireless systems, achieving high spectral efficiency by using large number of antennas at the base station (BS) [1]. This not only simplifies processing at the mobile user equipment with downlink precoding [2], but also provides good performance in the uplink with linear detection algorithms such as matched filtering (MF) and zero-forcing (ZF). However, the complexity of these algorithms increases linearly with the number of BS antennas (M), and quadratically with the number of active users (K). Furthermore, the propagation scenarios in a MaMi systems can vary more than traditional small-scale multiple-input multiple-output (MIMO) systems due to the distributed nature of several single antenna users. Hence, an adaptive signal detector which considers factors, such as the number of active users and channel conditions, is needed to achieve hardware and energy efficiency.

In this paper, a detector which employs MF detection for K_1 non-interference limited users close to the BS, and ZF detection for the other K_2 interference limited users is presented. The varying positions of mobile users over time, results in changing values of K_1 and K_2, which indicates the need for reconfigurable MF and ZF detectors. ZF detection, which is the more complex of the two, can be performed with Cholesky decomposition (CD) of the Gram matrix ($\boldsymbol{H}^H\boldsymbol{H}$) followed by forward backward substitution (FBS). Furthermore, the block CD algorithm can be used to construct the CD of larger matrices from the CD of smaller sub-matrices, enabling the implementation of a variable size CD processor. Nonetheless, CD has a complexity in the order of $\mathcal{O}(K^3)$, and if calculated on each tone in an orthogonal frequency division multiplexing (OFDM) system, a throughput proportional to the bandwidth is required. This traditional per-tone approach is neither scalable nor efficient for wide-band systems. Fortunately, channel conditions such as coherence bandwidth (B_{coh}) can be exploited to compute CD of a few selected tones and linear interpolation can be used to approximate CD of intermediate tones. This enables a drastic reduction in computation count, in proportion to the interpolation distance measured in tones (D_{sc}), at the cost of a slight increase in bit error rate (BER).

The proposed detection scheme with combined MF and CD based ZF is evaluated in a MaMi system with 128 antennas at the BS. Effects of CD interpolation are analyzed with measurement data from [3] and the hardware architecture for an interpolation based ZF detector is presented, reducing computations by up to 60 times. To reduce silicon area and power, the CD unit is optimized for 8 users and block decomposition is employed when all 16 users require ZF detection. The ZF detector is implemented in 28 nm FD-SOI with a 141 k-gates CD unit and has a peak throughput of 20 M CD/s. The adaptive interpolation feature is controlled by a single parameter D_{sc}, which simplifies hardware implementation and provides a wide tuning range.

II. Background

The baseband signal in the uplink of an OFDM system with N tones can be modeled as

$$\boldsymbol{y}_i = \boldsymbol{H}_i\boldsymbol{x}_i + \boldsymbol{n}_i, \qquad \forall i \in \{1, 2, \ldots, N\} \tag{1}$$

where, the subscript i is the tone index, $\boldsymbol{y}_i = [y_1, ..., y_M]^T$ is the received data, $\boldsymbol{H}_i \in \mathbb{C}^{M\times K}$ the channel gain matrix, $\boldsymbol{x}_i = [x_1, ..., x_K]^T$ the combined transmit data of users and $\boldsymbol{n}_i$ the additive white Gaussian noise. Signal detection with MF increases signal to noise ratio (SNR), providing good performance for non-interference limited user, and is achieved by

$$\widehat{\boldsymbol{x}}_{MF} = \boldsymbol{H}^H\boldsymbol{y} = \boldsymbol{H}^H\boldsymbol{H}\boldsymbol{x} + \boldsymbol{H}^H\boldsymbol{n}, \tag{2}$$

where $\boldsymbol{H}^H$ is the Hermitian conjugate of the channel estimate. However, in interference limited scenarios, the more advanced ZF detector is required which operates on the MF data by

$$\widehat{\boldsymbol{x}}_{ZF} = \left(\boldsymbol{H}^H\boldsymbol{H}\right)^{-1}\widehat{\boldsymbol{x}}_{MF}. \tag{3}$$

Typically, users are distributed over a cell, resulting in conditions where MF provides good BER performance for some users but others require ZF. This distribution varies over time due to mobility, sometimes leading to the worst case scenario where all users require ZF detection. Thus, reconfigurability in the signal detector for handling variable channel conditions is desired to achieve energy efficiency.

Operating channel conditions do not change drastically in low mobility scenarios. Therefore, in situations where the B_{coh} is in the order of several OFDM tones, different interpolation techniques can be used to lower ZF computations [4]. However, low complexity linear interpolation of the inverses with D_{sc} around 60 results in significant performance degradation.

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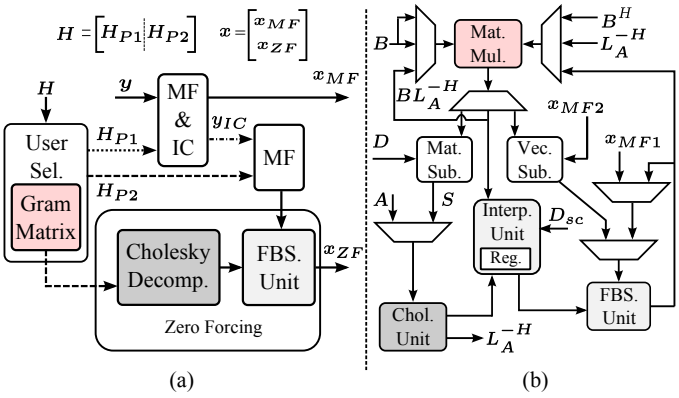
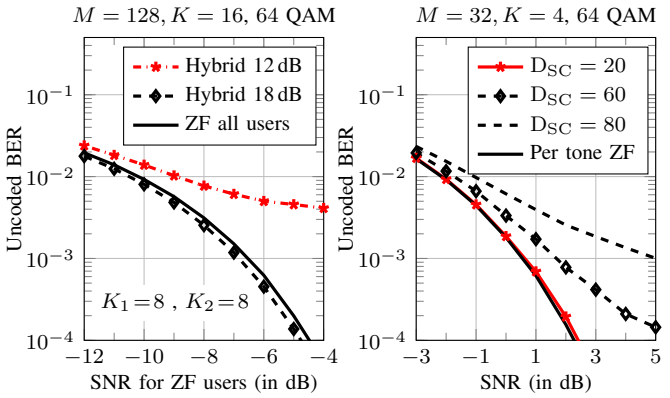


Fig. 1: (a) Signal flow in hybrid detector, (b) Block CD based ZF detector



(a) Hybrid detection with MF and ZF (b) Interpolation based CD

Fig. 2: BER simulations for hybrid detector and interpolating CD detector

III. Adaptive Detection with Interpolation

Even though a typical BS is equipped with a direct source of power, energy efficiency is important due to the large scale of MaMi systems. Hence, implementing a full ZF detector for the worst case scenario where all user are interference limited, will result in a suboptimal design. In the next section a hybrid detection scheme is analyzed which coupled with an interpolation strategy lowers complexity and power for minimal performance loss.

A. Hybrid detection with MF and ZF

From a power consumption perspective MF is preferred over ZF but results in inferior performance for interference limited users. To balance these conflicting demands of power and performance, a hybrid detector with user selection (User Sel.) capability is presented in Fig. 1 (a). The interference limited users are detected by comparing the diagonal and off-diagonal entries in the full Gram matrix. In limited mobility scenarios, where interference conditions do not change drastically, the computationally expensive full Gram matrix is calculated occasionally. The users with low interference are selected for MF with channel estimates $\boldsymbol{H}_{P1}$ followed by hard detection to produce $\boldsymbol{x}_{MF}$. Next, interference cancelation (IC) using $\boldsymbol{H}_{P1}$ and $\boldsymbol{x}_{MF}$ is done on the received data $\boldsymbol{y}$ to calculate a lower dimension vector $\boldsymbol{y}_{IC}$. The reduced Gram matrix for the remaining users is computed with $\boldsymbol{H}_{P2}$, followed by ZF detection of the interference free vector $\boldsymbol{y}_{IC}$, resulting in $\boldsymbol{x}_{ZF}$.

The average BER of 16 users in a system with 128 BS antennas operating with the proposed detection scheme is compared against full ZF of all users in Fig. 2 (a). In the simulations, eight users (K_1) with a higher SNR are detected with MF, whereas the other eight users (K_2) require ZF detection. When the SNR (per receive antenna) of K_1 is just 12 dB higher than K_2, significant degradation in BER is seen due strong interference, resulting in poor MF detection. However, when K_1 users have 18 dB or higher SNR, MF works well and the hybrid scheme performs slightly better than ZF on all users. This can be attributed to the smaller ZF matrix size ($K_2 = 8$) and the resulting lower noise enhancement in the hybrid scheme. Another advantage of the proposed scheme is the reduced dynamic range of the Gram matrix entries generated from $\boldsymbol{H}_{P2}$, as only users with similar SNR are chosen for ZF. Thus, the hardware implementation of ZF detector in the hybrid scheme can be optimized to use lower precision when compared to a ZF detector for all users.

B. Adaptive Interpolation of Cholesky Decomposition

The proposed hybrid scheme reduces the number of users requiring ZF detection and hence the computation count when some users are not limited by interference. Further reductions are obtained by exploiting the B_{coh} of the channel. Instead of direct matrix inversion of Gram matrices at each OFDM tone, the Gram matrix inverse of two tones which are at a distance of D_{sc} apart are interpolated for the intermediate tones. However, linear interpolation of these inverses causes significant degradation in performance [4]. As an alternative, consider ZF detection based on CD of the Gram matrix with

$$\widehat{\boldsymbol{x}}_{ZF} = \left(\boldsymbol{H}^H\boldsymbol{H}\right)^{-1}\widehat{\boldsymbol{x}}_{MF} = \left(\boldsymbol{L}\boldsymbol{L}^H\right)^{-1}\widehat{\boldsymbol{x}}_{MF}, \tag{4}$$

followed by forward backward substitution (FBS) with,

$$\begin{aligned}\boldsymbol{y}_F &= \boldsymbol{L}^{-1}\widehat{\boldsymbol{x}}_{MF}\\ \widehat{\boldsymbol{x}}_{ZF} &= \boldsymbol{L}^{-H}\boldsymbol{y}_F,\end{aligned} \tag{5}$$

where $\boldsymbol{L}$ is a lower triangular matrix. To lower complexity, linear interpolation of the $\boldsymbol{L}$ matrices is proposed instead of Gram matrix inverse interpolation. Furthermore, the interpolation distance D_{sc} is adaptively varied based on the B_{coh} of the operating channel to maintain low BER loss, similar to the method considered in [5] for reducing QR decomposition (QRD) computation cost. If $\boldsymbol{L}_1$ and $\boldsymbol{L}_N$ correspond to the CD components from tones which are D_{sc} apart, then the linear interpolated $\boldsymbol{L}_i$ of intermediate tones are computed with a factor α dependent on D_{sc} using,

$$\boldsymbol{L}_i = (1-\alpha)\boldsymbol{L}_1 + \alpha\boldsymbol{L}_N = \boldsymbol{L}_1 + \alpha(\boldsymbol{L}_N - \boldsymbol{L}_1). \tag{6}$$

The error introduced by this approximation is reduced by decreasing D_{sc} whereas computation count is reduced by increasing D_{sc}, enabling run time trade-off between power and performance. The effects of CD interpolation on BER in a MaMi system are evaluated with actual channel measurement data from [3]. A tone spacing of 15 kHz is used with different values of D_{sc} and the resulting BER is depicted in Fig. 2(b), showing a loss of less than 1 dB at 10^{-3} for D_{sc} as high as 60. This highlights the potential of replacing CD computations by low complexity linear interpolation from (6).

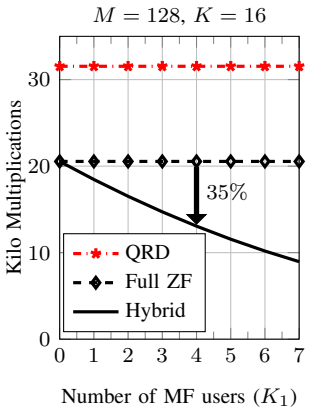
C. Complexity analysis

The comparison of three linear detection methods with multiplications as the metric is shown in Fig. 3 (a). QRD based detection is often used in small-scale MIMO systems but is

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Operation	Multiplications
QRD based	
QRD [6]	$MK^2 - K^3/3$
Back Substitution	$K(K+1)/2$
Full ZF	
Gram Matrix	$MK(K+1)/2$
MF	MK
CD based ZF	$K(K+1)(K+2)/6$
FBS	$K(K+1)$
Proposed Hybrid	
User Sel.	MK
MF	MK
IC.	~ 0
Gram Matrix	$MK_2(K_2-1)/2$
Partial CD	$K_2(K_2+1)(K_2+2)/6$
FBS	$K_2(K_2+1)$
Interpolation (6)	$K_2(K_2+1)/2$

(a) Comparison of detection schemes



(b) Operation count comparison

Fig. 3: Complexity analysis of detection schemes

not beneficial in terms of complexity in MaMi systems. The CD based full ZF detection has lower storage requirements and complexity than QRD but the proposed hybrid scheme requires the smallest number of multiplications. Fig. 3 (b) shows the significant reduction obtained with the hybrid scheme when the number of MF users (K_1) increases. Further reductions by a factor of $(K_2+2)/3$ is obtained by replacing CD with interpolation from (6). As an example, in a system with $M = 128$, $K = 16$ and $K_1 = 4$, the hybrid detection scheme reduces complexity by 35%.

IV. VLSI ARCHITECTURE

The top level signal flow diagram for the proposed hybrid detector is shown in Fig. 1 (a). The received data $\boldsymbol{y}$ and the columns $\boldsymbol{H}_{P1}$ corresponding to users selected for MF are forwarded to the MF and IC unit. The estimates $\boldsymbol{H}_{P2}$ for K_2 users are used to compute the reduced Gram matrix for ZF detection. In the following, optimizations for the ZF detector which consists of a CD unit and a FBS unit are discussed.

A. Block Based Cholesky decomposition

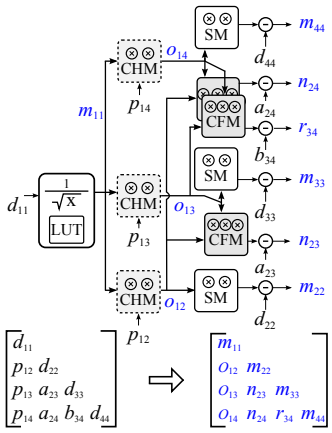
The hybrid detection scheme requires a reconfigurable CD unit, for detecting variable number of users with ZF, depending on interference conditions. A systolic array based CD unit can be obtained by modifying the design in [7] which has around 60% multiplier utilization. The current implementation is based on a similar structure and the internal connections with input-output matrices for a part of the CD unit are shown in Fig .4 (a). A combination of different types of multipliers for scaling (CHM), finding the absolute value (SM) and for full complex multiplication (CFM) are used. To save silicon area, the CD unit is optimized for 8 users and block CD algorithm is used to handle scenarios where more than 8 users need ZF detection. Consider the Gram matrix for 16 users

$$\boldsymbol{H}^H\boldsymbol{H} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}^H \\ \boldsymbol{B} & \boldsymbol{D} \end{bmatrix} \text{ and } \boldsymbol{S} = \boldsymbol{D} - \boldsymbol{B}\boldsymbol{A}^{-1}\boldsymbol{B}^H, \quad (7)$$

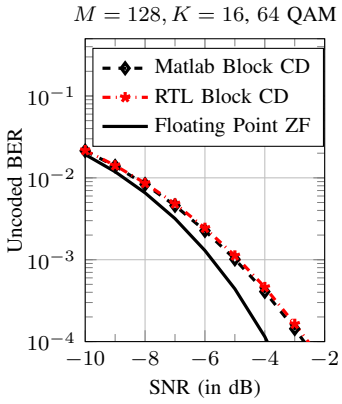
the Schur complement obtained with smaller 8×8 submatrices $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{D}$. With block decomposition algorithm, the

$$\text{CD}(\boldsymbol{H}^H\boldsymbol{H}) = \begin{bmatrix} \boldsymbol{L_A} & \boldsymbol{0} \\ \boldsymbol{B}\boldsymbol{L_A^{-H}} & \boldsymbol{L_S} \end{bmatrix}, \quad (8)$$

where $\boldsymbol{L_A}$ and $\boldsymbol{L_S}$ are the CD components of $\boldsymbol{A}$ and $\boldsymbol{S}$ respectively. This method, requires $\boldsymbol{A}^{-1}$, which in a MaMi



(a) Internal connections in part of CD unit with an example transformation



(b) BER of Block CD and RTL

Fig. 4: Architecture of CD unit and performance comparison of Block CD

system can be approximated by

$$\boldsymbol{A}^{-1} \approx \mathbb{E}\left[a_{jj}^{-1}\right] = \gamma, \text{and } \boldsymbol{S} \approx \boldsymbol{D} - \gamma\boldsymbol{B}\boldsymbol{B}^H, \quad (9)$$

where a_{jj} are the diagonal elements of $\boldsymbol{A}$ and the inversion operation of a_{jj} is obtained with a look up table (LUT). The $\boldsymbol{B}\boldsymbol{L_A^{-H}}$ product is computed with the special $\boldsymbol{L}_{A_j}^{-H}$ matrices [6] generated during the CD of $\boldsymbol{A}$ from

$$\boldsymbol{B}\boldsymbol{L}_A^{-H} = \boldsymbol{B}\left(\boldsymbol{L}_{A_1}^{-H}\boldsymbol{L}_{A_2}^{-H}\ldots\boldsymbol{L}_{A_j}^{-H}\right), \forall j \in \{1,\ldots,8\}, \quad (10)$$

which by construction modify only the corresponding j-th column of $\boldsymbol{B}$. This operation has the same complexity and structure as $\boldsymbol{B}\boldsymbol{B}^H$ in (9) and can be computed by the same hardware unit. Furthermore, these operations can be shared with the Gram matrix unit. Fig. 1 (b) shows the connections in the block CD implementation, using a 8×8 CD unit, a FBS unit together with a matrix multiplier (from Gram matrix unit). Fig. 4 (b) shows the BER performance of the block CD based detector, indicating a loss of less than 1 dB at 10^{-3} when compared to floating point ZF. The timing schedule in Fig. 5 describes the execution order for 16 users, where the matrix multiplier operates in three modes, producing the Schur complement, the CD component $\boldsymbol{B}\boldsymbol{L_A^{-H}}$, and performing parts of forward substitution (Vec). When detecting 8 or lower number of users, only the CD, interpolation (IU) unit and FBS are active and if 12 users require ZF, block decomposition with zero padding is used.

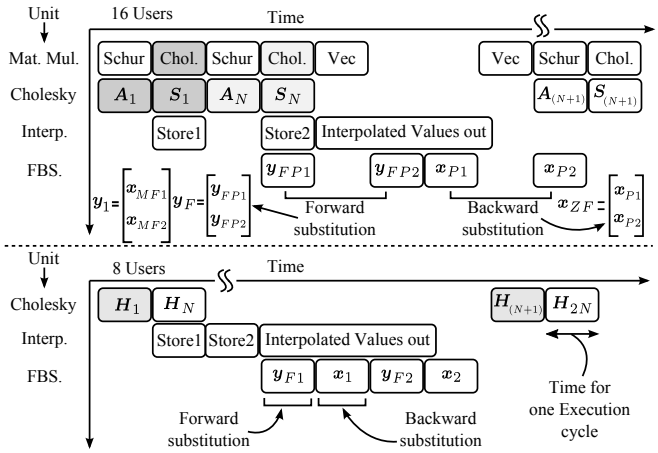


Fig. 5: Timing schedule for $K = 16$ and $K = 8$

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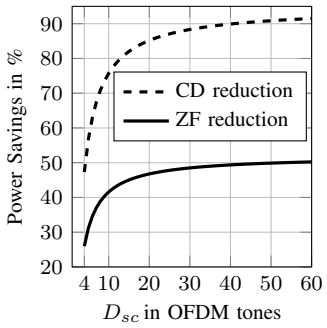


TABLE I: Implementation results

Unit	Users (K_2)	Rate[1]	Gate count	Power (mW)[2]
CD	8	20 M	141 k	**28**
	4			**20**
FBS	8	40 M	95 k	23
	4			19
IU	8	20 M	45 k	**1.5**
	4			**1**

[1] Effective throughput is halved when $K_2 = 16$ due to block decomposition.
[2] Post layout simulation.

Fig. 6: Power reduction with adaptive ZF detector and implementation details

B. Interpolation unit and Forward Backward Substitution unit

The IU unit stores two consecutive output $\boldsymbol{L}_1$ and $\boldsymbol{L}_N$ from the CD unit in the "Store1" and "Store2" cycles as shown in Fig. 5 and produces the $\boldsymbol{L}_i$ matrices using (6) for the intermediate tones. The parameter D_{sc} determines the α weights and the unit is implemented to produce a new $\boldsymbol{L}_i$ every two FBS runs. The FBS unit operates on $\boldsymbol{y}_{IC}$ with the $\boldsymbol{L}_i$ matrix from the IU unit and runs twice for each input $\boldsymbol{x}_{MF}$, performing forward followed by backward substitution. However, when 16 user ZF detection is required, the block is run 4 times as depicted in Fig. 5. The vector $\boldsymbol{y}_{FP1}$ from the forward substitution of $\boldsymbol{x}_{MF1}$, is multiplied with $\boldsymbol{B}\boldsymbol{L}_{\boldsymbol{A}}^{-\boldsymbol{H}}$, subtracted from $\boldsymbol{x}_{MF2}$ using the Vec. Sub. unit followed by the second forward substitution to produce $\boldsymbol{y}_{FP2}$. The same procedure is repeated to obtain the final result $\boldsymbol{x}_{ZF}$.

V. Implementation and Results

The current implementation adopts a high level synthesis [8] flow which enables rapid design space exploration to optimize hardware reuse. The inputs and internal variables of the CD unit are quantized to 13 bits and a target throughput of 20 M CD/s is chosen to support wide-band systems and full ZF detection without interpolation. The CD unit for 8×8 matrices requires 120 (most of them complex) multiplication operations corresponding to 288 real multipliers. To save silicon area, the current design uses a folding factor (FF) of 20 and a clock of 400 MHz resulting in a implementation with 19 real multipliers operating at 80% utilization. In order to match the throughputs, the FBS unit is implemented with a clock of 600 MHz and a FF of 15. The operations performed in the FBS unit are sensitive to division accuracy and a high precision unit is implemented and reused. When the ZF detector operates in the interpolation mode, the IU unit clocked at 40 MHz generates a new output $\boldsymbol{L}_i$ every 2 clock cycles to be used in the FBS unit.

The BER performance of the implementation is compared with floating point models in Fig. 4 (b), indicating a marginal increase. The components of the ZF detector are synthesized and implemented in 28 nm FD-SOI technology and the implementation details are shown in Table I. Though the addition of the IU unit adds 19% area overhead, switching to interpolation mode allows CDs to be replaced by low power interpolations, e.g., with a D_{sc} of 20, the detector would compute 2 CDs and 18 interpolations instead of 20 CDs. The reduction in power dissipation for different values of D_{sc} is depicted in Fig . 6 for both CD and ZF detector (includes FBS). A comparison of the CD unit with other implementations from literature is presented in Table II. Compared to small-scale MIMO decomposition implementations in [9] [10], the proposed design has higher energy efficiency with comparable gate efficiency, and can be further improved by using adaptive interpolation strategy.

TABLE II: Comparison of Decomposition Processors

Item	NS [2]	LDL [9]	LU [10]	This work (CD)
Matrix Dimension (N)	16×16	4×4	4×4	8×8
Technology [nm]	65	90	90	28
Gate Count [k]	104	90	68	141
Throughput [M ops/s]	0.5	30	31.5	20
Norm. Throughput [M ops/s] ⋆	8	16	17	20
Norm. Gate Efficiency †	77	178	250	**141**
Power [mW]	-	-	35◇	**28**
Norm. Energy‡ [nJ]	-	-	3◇	**1.4**

⋆ Normalized (Norm.) throughput = Throughput $\times \frac{\text{Technology}}{28} \times \frac{(N)(N+1)(N+2)}{(8)(9)(10)}$
† Norm. Throughput / Gate Count
◇ Measurement result (includes forward substitution)
‡ Norm. Energy = Energy per matrix operation $\times \frac{(8)(9)(10)}{(N)(N+1)(N+2)} \times \left(\frac{0.8}{V_{dd}}\right)^2$

VI. Conclusion

This paper presents an adaptive uplink detector for MaMi systems by using a hybrid detection scheme. A CD interpolation strategy exploiting B_{coh} is employed to lower computational complexity and is verified with measured channel data. A reconfigurable architecture based on CD unit and block decomposition serving up to 16 users is presented. The ZF detector is implemented in 28 nm FD-SOI technology, has a peak throughput of 20 M CD/s and requires 1.4 nJ/CD. The CD interpolation scheme reduces power dissipation of the ZF detector by 50% with a performance loss of less than 1 dB, resulting in a highly energy efficient signal detector.

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